

End Semester Examination
School of Computer and Systems Sciences
Jawaharlal Nehru University
MCA III Semester
Cloud Computing and IoT (CS -426)

December 16, 2023
MM: 50
Time: 3 hrs.

- Q1.** In the context of Cloud computing, differentiate between *Scale Up* and *Scale Out*. (3)
- Q2.** In the context of Multiprocessors discuss the advantages of using Physical partitioning. (3)
- Q3.** How addressing scheme can be implemented in IoT based systems? (3)
- Q4.** What is the significance of CAP Theorem? Discuss it in the context of Cassandra Key-Value storage. (3)
- Q5.** Discuss the use of Compaction and Tombstone in the context of Cassandra. (1.5*2=3)
- Q6.** In the context of Cloud Computing explain the following properties of virtualization.
- a) Encapsulation
 - b) Interposition
- (2.5*2=5)
- Q7.** Draw and discuss a typical Entity based IoT Architecture (5)
- Q8.** Compare the Dynamic provisioning schemes in the context of Reservation based scheme and Spot pricing scheme. (5)
- Q9.** What is Critical Section in a program code? Draw a figure to present the working of a Decode and Dispatch Interpreter. Discuss the problems associated with the same. (1+2+2=5)
- Q10.** Discuss the four most important properties of a failure detector. Discuss the same in the context of an All to All Heart beating based Failure Detector. (2+3=5)
- Q11.** For Scalable Weakly-consistent Infection Style Membership Protocol (SWIM) failure detector, prove that for its any process, Probability of being Pinged in Time $T = 1 - e^{-1}$ meaning that there is a certainty in pinging all the processes for failure detection. (5)
- Q12.** Let the availability of a compute cluster be 99.99% with a planned downtime of 52 hours. Calculate the effective availability of the system. (5)

***** All the Best*****